

Kiyan Amirian

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Research Summary

PhD student in Materials Science and Engineering at the University of Maryland. Developed FP-DeErr, an open-source framework and public leaderboard benchmarking foundation potentials (universal MLIPs) against DFT on force prediction, phase stability, elemental ordering and ion migration, exposing failure modes that average errors hide. Experience spans atomistic simulation (DFT, molecular dynamics, NEB), hands-on LLM implementation and Python research software development, orchestrating coding agents (Claude Code and Codex). Available for internship: Oct 2026 - Aug 2027 (flexible).

Education

University of Maryland

Doctor of Philosophy in Materials Science and Engineering
Cumulative GPA: 4.0/4.0

College Park, MD, USA

2023-Expected Aug 2027

Shiraz University

Bachelor of Science in Materials Science and Engineering
Cumulative GPA: 17.15/20 (3.5/4.0); Last 2-year GPA: 18.54/20 (3.9/4.0)

Shiraz, Fars, Iran

2018-2023

Research Experience

Graduate Researcher, UMD (Advisor: Prof. Yifei Mo)

2024-Present

- **FP-DeErr:** Developed an open-source, application-oriented framework that decomposes foundation-potential (FP) errors by the physical quantities, configurations and workflow stages governing each task. Built its [evaluation code](#) and [public leaderboard](#), which benchmarks 18+ FPs against DFT (ASE, MatCalc, pymatgen).
- **Benchmark automation:** Implemented automated job generation and high-throughput benchmark execution using SLURM job arrays. Released [DFT-NEB reference data](#) and [chalcogenide benchmark subsets](#) on Figshare.
- **Force prediction:** Decomposed force errors into magnitude and angle and designed metrics for highly accurate predictions, large-force-error atoms and far-from-equilibrium atoms (MatPES, OMat24), revealing failure modes obscured by average MAE/RMSE ([GitHub](#), [leaderboard](#)).
- **Phase stability and elemental ordering:** Designed convex-hull agreement metrics (ground state, within-phase and global hull minimum) and assessed elemental-ordering rankings using Top-1, Recall@ k and Spearman's ρ on chalcogenide phase-change materials, separating intrinsic energy-ranking errors from FP-relaxation errors ([GitHub](#), [leaderboard](#)).
- **Ion migration:** Developed a workflow to benchmark FP-NEB against DFT-NEB on 154 Li- and Na-ion pathways, with metrics for convergence, profile shape, endpoint energy ranking and barriers, tracing barrier errors to intrinsic PES error, endpoint relaxation and along-path forces ([GitHub](#), [leaderboard](#)).

Undergraduate Researcher, Shiraz University (Advisor: Prof. Ramin Ebrahimi)

2019-2022

- Developed an algorithm to predict tensile instability in functionally graded low-carbon steel and compared its predictions with published experimental data ([paper](#), [code](#)).

Selected Machine Learning Projects

KV-Cache Optimization for Large Language Models

Fall 2025

Led a team comparing KV-cache compression techniques (StreamingLLM, KIVI, ZipCache) for long-context LLM inference. Reimplemented KIVI 2-bit quantization for LLaMA-2 7B/13B using PyTorch. Designed evaluation workflows across summarization, reasoning, and QA benchmarks, adding metrics for token match rate, entity hallucination, and sentence compliance ([GitHub](#)).

Image Classification and Transfer Learning

Spring 2025

Implemented logistic regression from scratch in NumPy; used PCA with an RBF SVM for MNIST (97.1% test accuracy) and fine-tuned VGG19 for a 10-class monkey-species dataset (97.06% validation accuracy) ([GitHub](#)).

Face Recognition and Kernel Methods

Spring 2025

Implemented a custom gradient-descent kernel SVM to address CVXOPT numerical failures with high-degree polynomial kernels. Benchmarked PCA/MDA with Bayes, k-NN, SVM, and AdaBoost on a 200-subject dataset ([GitHub](#)).

Publications and Manuscripts

K. Amirian, R. S. Saravanan, F. Adams, C. E. Schwarz, Y. Mo

arXiv, 2026

FP-DeErr: Application-Oriented Error Decomposition for Foundation Potentials

[arXiv:2609.05714](#) · [GitHub](#) · [Leaderboard](#)

R. S. Saravanan, C. E. Schwarz, F. Adams, K. Amirian, Y. Mo

Submitted

A Simple Universal Descriptor for Solid-State Ionic Conductors Across Chemistries

K. Amirian, Z. Abbasi, R. Ebrahimi

2023

Analytical and numerical instability analysis of functionally graded low-carbon steel. International Journal of Iron and Steel Society of Iran, 19(2), 66–72. [doi:10.22034/IJISSI.2023.1990580.1262](#)

Skills

Programming: Python (NumPy, pandas, SciPy, Matplotlib), Bash, MATLAB, Fortran, Git, $\LaTeX$

Atomistic Simulation: DFT (VASP), LAMMPS, ASE, pymatgen, MatCalc; NEB, molecular dynamics

Foundation Potentials (MLIPs): MACE, CHGNet, M3GNet, UMA, Orb, MatterSim, DPA4, eSEN

ML Frameworks: PyTorch, Hugging Face Transformers, TensorFlow/Keras, scikit-learn

ML Methods: LLM KV-cache quantization, transfer learning, CNNs, SVMs

Continuum Modeling: Abaqus (FEA)

Coding Agents: Claude Code, Codex

HPC: SLURM

Honors & Awards

Dean's Fellowship Award, University of Maryland, \$43,000

Spring 2023

Ranked 2nd of 56, B.Sc. Materials Science and Engineering, Shiraz University

Spring 2022

Relevant Graduate Courses

Introduction to Atomistic Modeling (4.0/4.0)

Natural Language Processing (4.0/4.0)

Statistical Pattern Recognition (4.0/4.0)

Machine Learning for Materials Science (4.0/4.0)

Teaching Experience

TA: Kinetics, Diffusion and Phase Transformations (Fall 2024)

TA: Mechanical Properties II (Spring 2022)

TA: Fundamentals of Computer (Spring 2020)

High-school algebra instructor (Fall 2019)

Research Interests

Computational Materials Science

Machine-Learned Interatomic Potentials

Atomistic Simulation

AI for Science

References

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